Funded by

TechForChange

COHORT 4









VIISAR

Turning STEM into Touch.

Funded by ciena | nasscom foundation

Adarsh Goyal

Founder & CEO

Sneha Gupta

Co-Founder & COO

Naman Dadhich

Web Developer

Sarthak Kakar

Web Designer

PROBLEM STATEMENT

Funded by ciena | nasscom foundation

01.

Science and maths concept difficult to understand in braille.

02.

Insufficient Braille/ Educational Resources for Science and Maths subjects

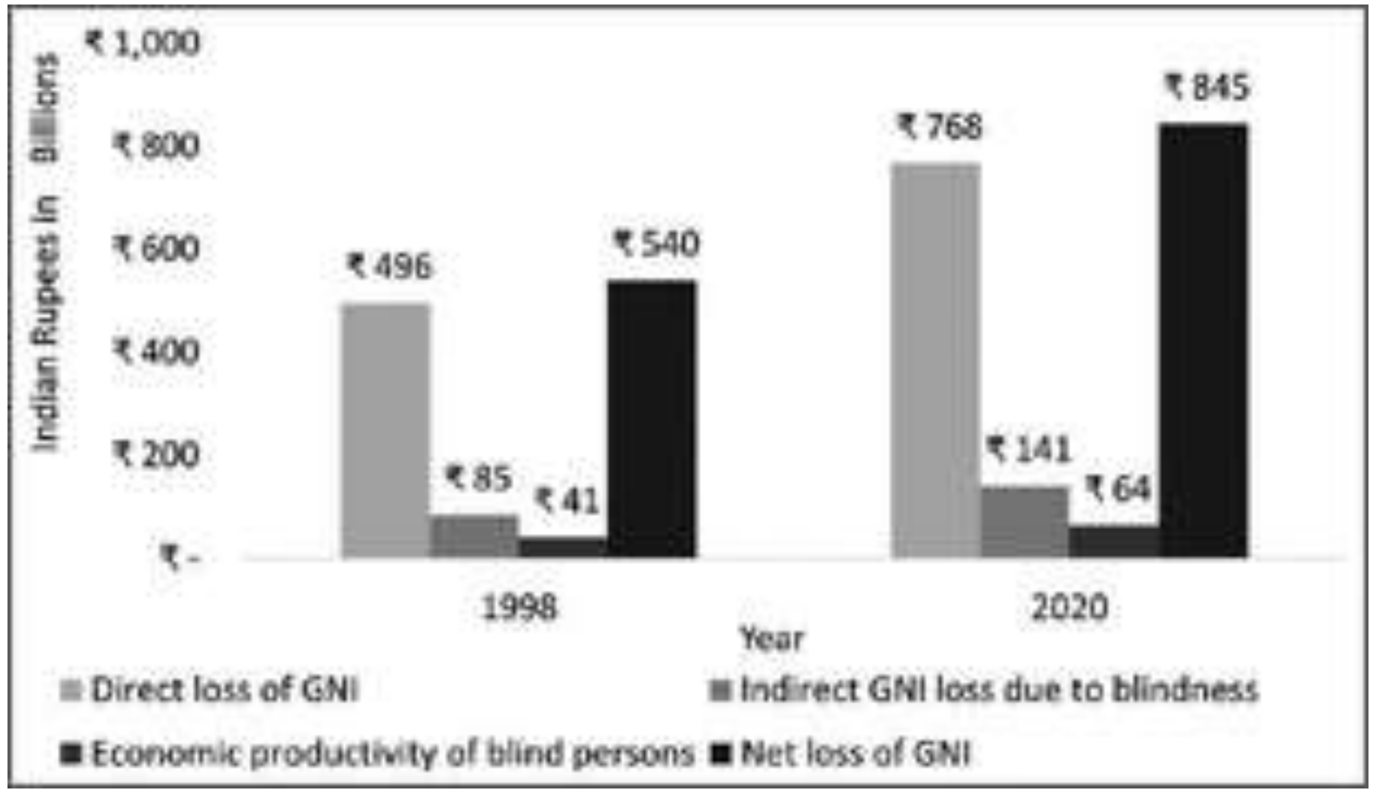
03.

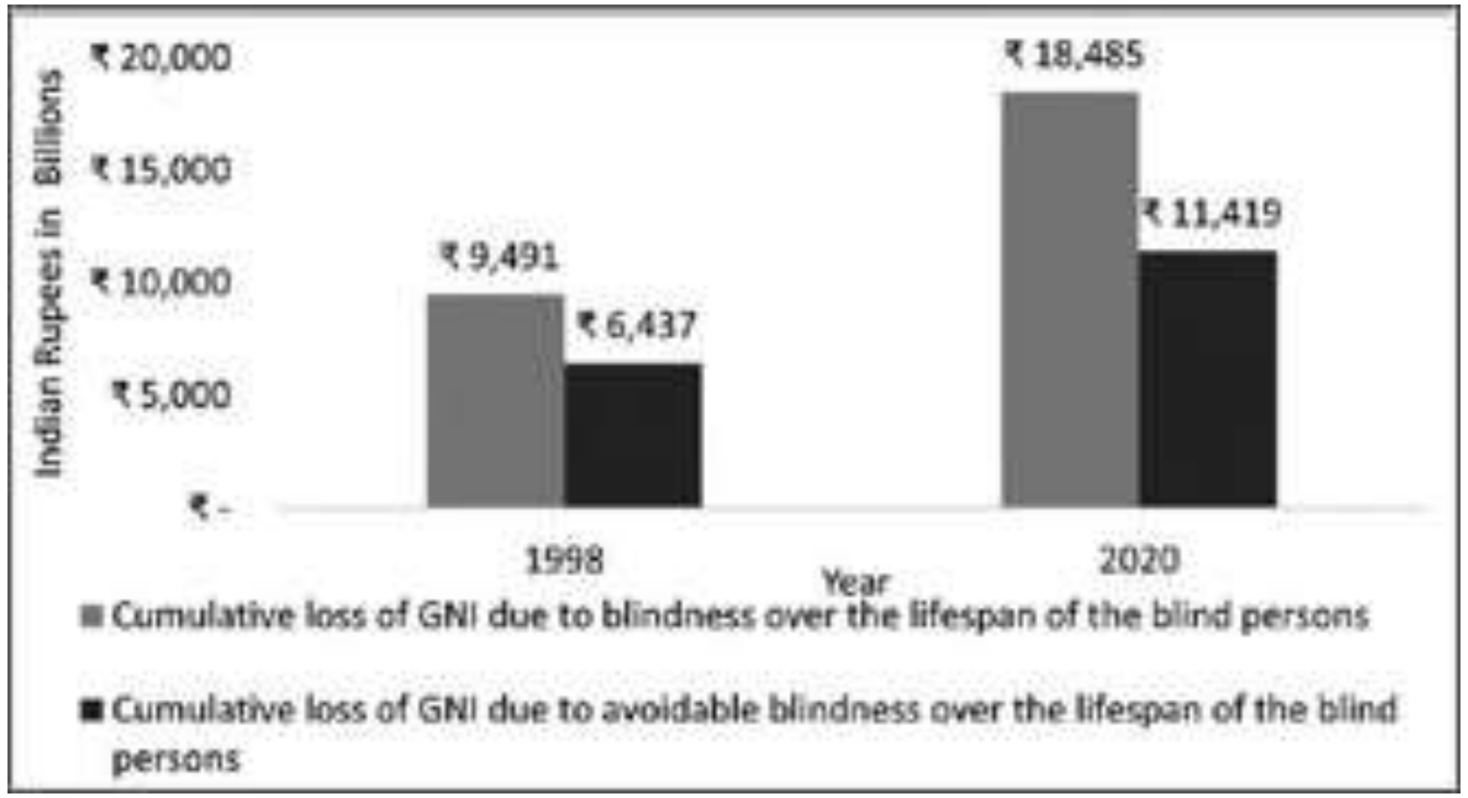
Learning Maths & Science from audio is difficult.

04.

Blind education is expensive

How Money is Lost?





How Time is Lost?

Table-3: Problems faced by visually impaired students

Statement		The most	More	Less Urgent	Absolutely not needed	Do not Know	χ^2	P value
Unavailability of information	Number	7	10	6	3	0	3.84	.279
	%	26.90%	38.50%	23.10%	11.50%	0.00%		
It takes more time to get information	Number	8	10	8	0	0	.308	.857
	%	30.80%	38.50%	30.80%	0.00%	0.00%		
Resources not available	Number	6	13	7	0	0	3.30	.191
	%	23.10%	50.00%	26.90%	0.00%	0.00%		
Don't Know , how to use resources	Number	6	4	13	3	0	9.38	.025
	%	23.10%	15.40%	50.00%	11.50%	0.00%		

VIISAR

Research & Market Analysis

Funded by ciena | nasscom foundation

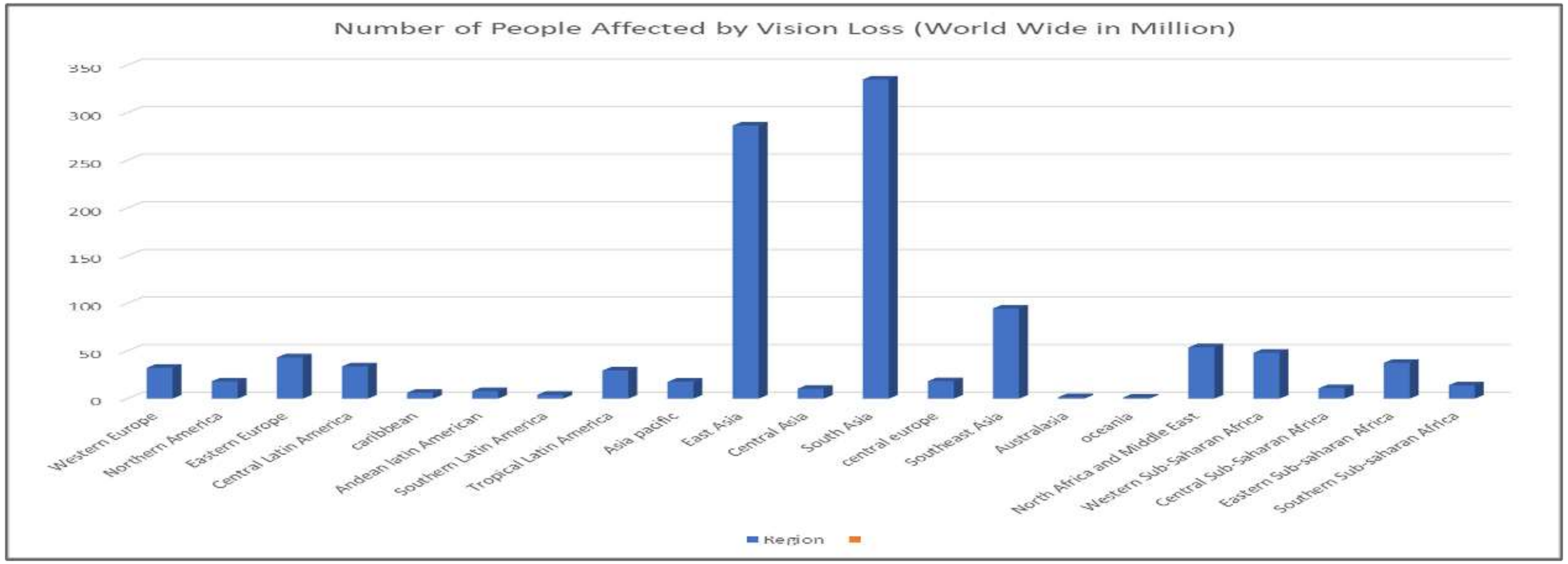
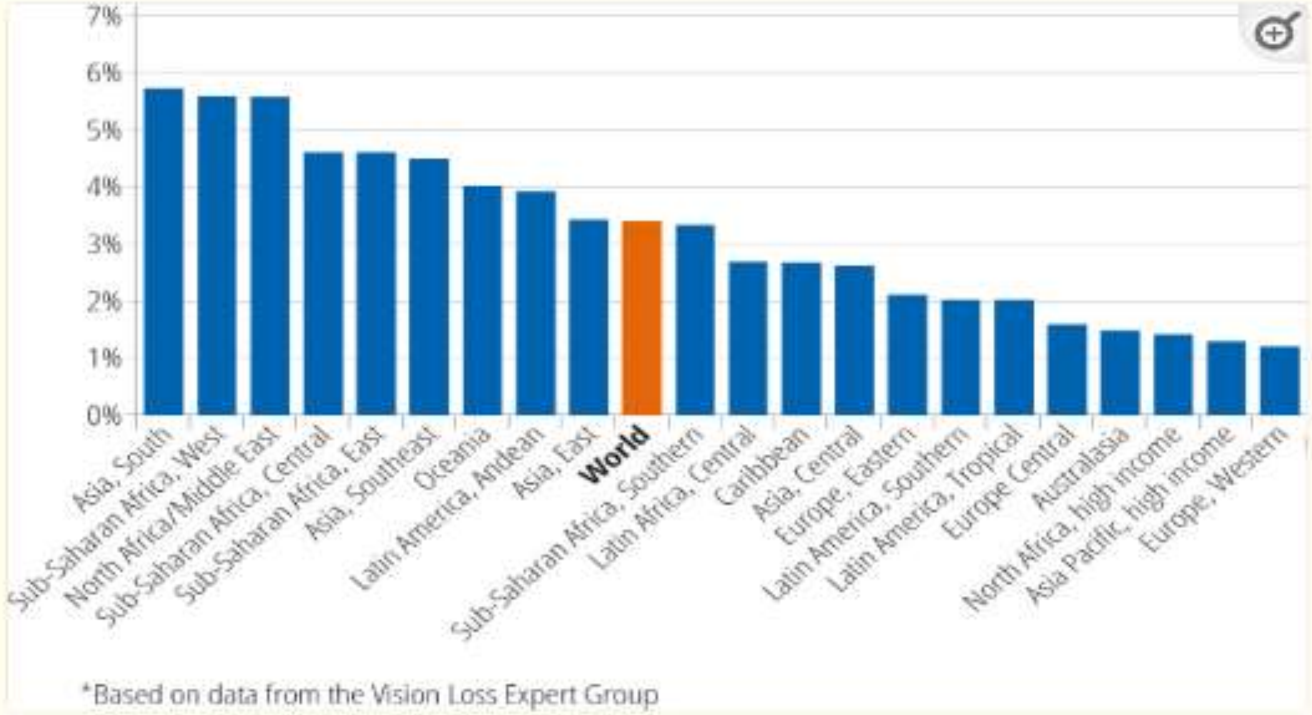


Figure 3

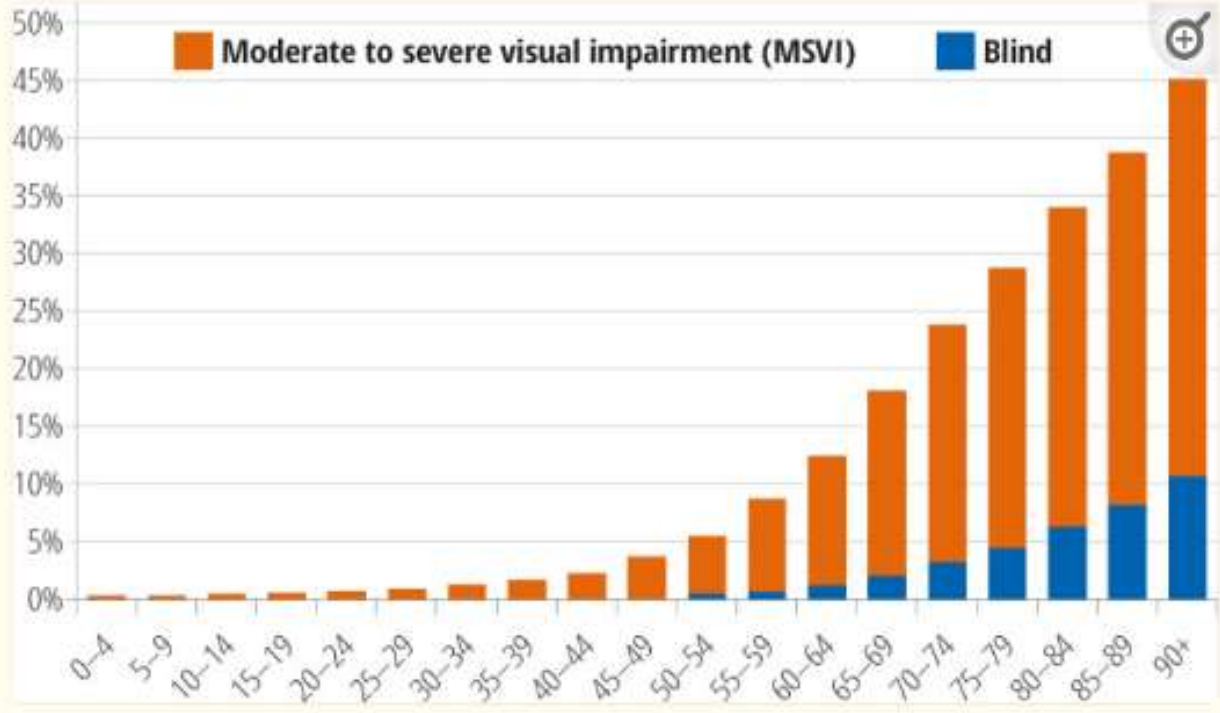


*Based on data from the Vision Loss Expert Group

Age-standardised prevalence of visual impairment across the 21 regions*

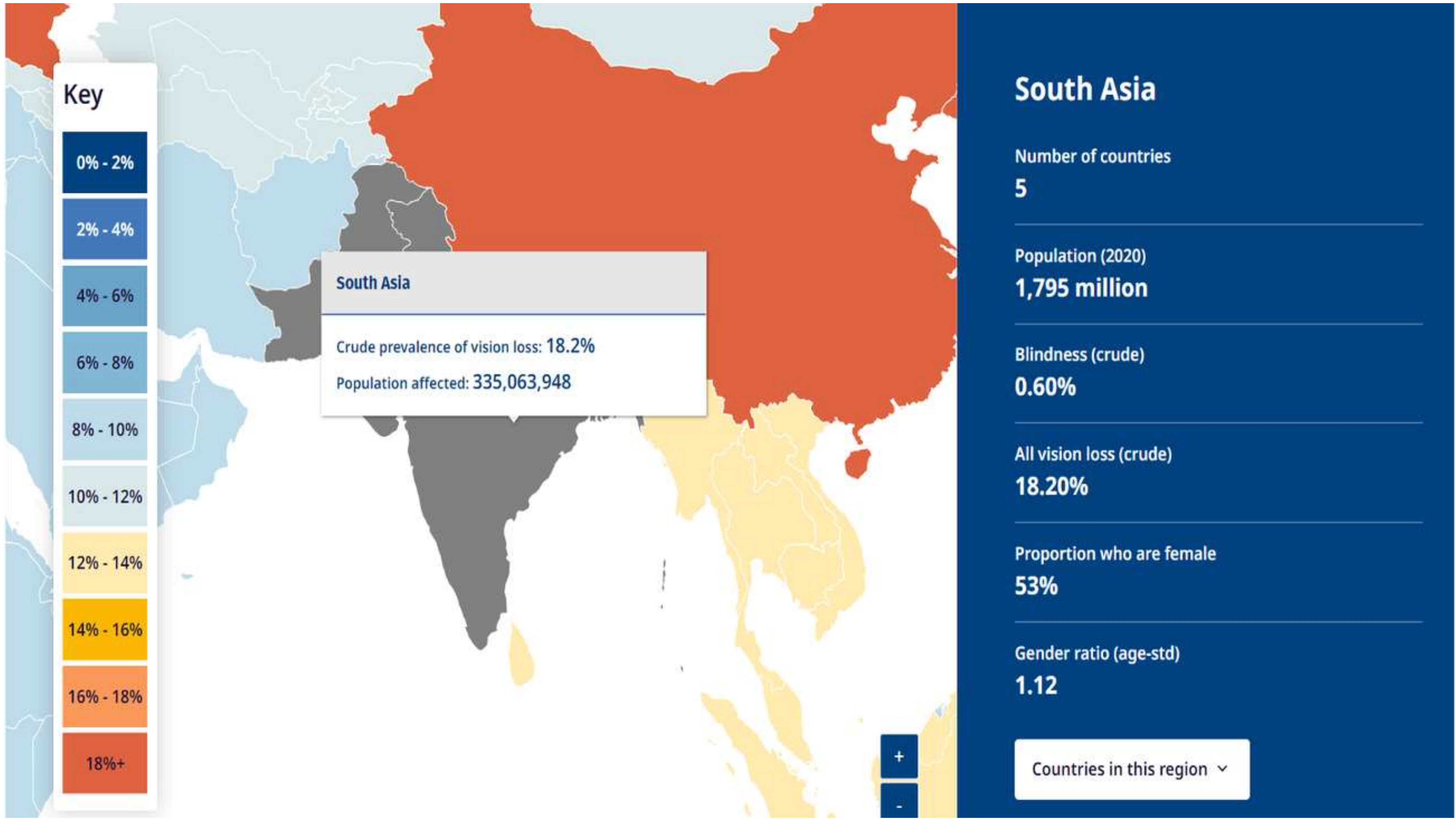
*Based on data from the Vision Loss Expert Group

Figure 2



The global prevalence of blindness and moderate to severe visual impairment (MSVI) in women*

Age	India	Worldwide
Preschool age (0–6 years)	~0.42 million	~31.8 million
School-age (5–19 years)	~1.18 million	~67.4 million
India – College-age / young adults (20–24 years):	~0.39 million	~20.6million



Braille literacy is low around the globe. In the United Kingdom, a more developed nation than India, the literacy rate for blind people is only 4%. The low number correlates with low employment rates for blind people. Existing methods of teaching are part of the reason the number is so low; it takes one student one

BORGEN Magazine

THE GOOD NEWS | BUSINESS & NEW MARKETS | POLITICS | CELEB | TECH | WORLD

HUMANITY, POLITICS & YOU

FIGHTING THE DENIAL OF EDUCATION FOR BLIND PEOPLE IN INDIA

Vasudha M, a tailor and mother of a blind daughter exclaimed: Most special educators and general teachers in these integrated schools themselves aren't adept in Braille. She struggles to read the texts. **Although there are assistive devices, they aren't accessible or available to everyone.** In Tamil Nadu, **government schools provide study materials in Braille only up to Class 8.**

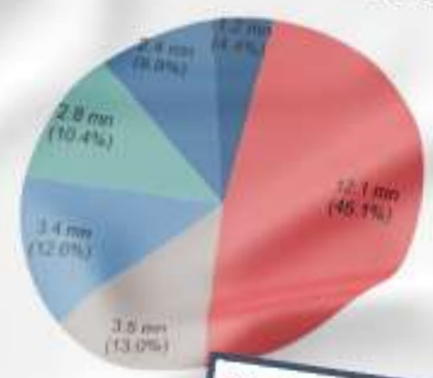
If inadequacy is an issue, the increasing price of Braille books also burdens the community. The price of the books is over 30-40 per cent higher than regular editions due to the high cost of printing.

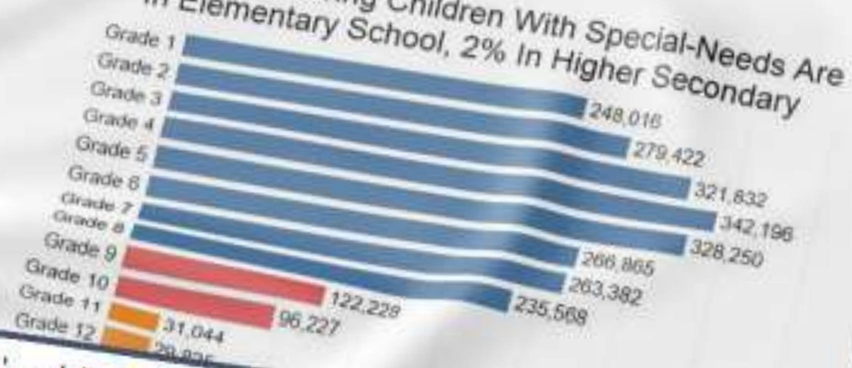
Muniyappan, a History professor and member of the College Students and Graduates Association for the Blind (CSGAB), explains how a lack of Braille books can determine a visually impaired person's future.

When we lack Braille books for school-level subjects, imagine wanting to study Engineering but **having to learn the voluminous syllabus through audio lessons**. That **is why most students take up Arts and Humanities and** do not **branch out to other professional streams.** The failure to provide access creates barriers for education and employment, and even undermines an individual's capability," he elucidates.

One education policy does not fit all

As things stand, 45% of India's disabled population is illiterate, according to Census 2011, compared to 26% of all Indians. Of persons with disabilit[ies] who are educated, 59% complete Class X, compared to 67% of the genera[l] population.





Six dots and social inclusion

She opens her eyes and looks at her daughter, who is now intently focusing on her mother's voice.

At least two visually-challenged students have been in the news recently for approaching the courts – Kritika Purohit filed a case in the Bombay High Court to be allowed to study physiotherapy and Reshma Dileep approached the Kerala High Court to be allowed to study science beyond secondary school.

Of the world's 37 million blind people, an estimated 90% live in developing countries. About 15 million of them live in India, a number that has doubled since 2007 to make it the highest number in any country. However, the braille literacy rate there is only 1%, far lower than the regular literacy rate of 77.7%. This

VIISAR

Funded by ciena | nasscom foundation

User interviews/survey insights

Click Here - https://www.youtube.com/watch?v=P_1gWaKteHY

User Validation & Confirmation

Funded by ciena | nasscom foundation

Target Users:

Wevisited **ShriSai Nethraheen Sanstha** and **Blind Person Association** in Uttam Nagar and we tested our product on visually impaired individuals.

25+ Users Tested

Feedbacks Received Based on Interviews and Testing Sessions:

Right now we mostly rely on audiobooks, but they don't help us feel shapes. Geometry books and physical models are also heavy and hard to carry. Your device solves this problem because we can feel different shapes on a single device, and it is much easier to carry.

Rajesh

This product helps us imagine shapes in our mind. For ex- it helps us understand how A looks, how a triangle feels, or what other figures are like. For blind students, it is very helpful because it allows us to connect more with what we are studying. With this device, understanding shapes and figures becomes much easier.

Krishna

The device feels a bit large right now. If the size could be reduced, it would be better because then we could feel the whole image using just our palm. That would make it more comfortable and easier to use.

Rajesh

Video of the MVP

Funded by

[Click Here - https://www.youtube.com/watch?v=P_1gWaKteHY](https://www.youtube.com/watch?v=P_1gWaKteHY)

VIISAR

Current Solutions vs Existing Gaps

Funded by ciena | nasscom foundation

Factor	Graphiti	Dot Pad	BrailleMe
Cost accessibility	Low	Low	High
Multi-line support	Yes	Yes	No, mainly 20-cell single-line
STEM suitability	Strong but expensive	Strong but expensive	Limited
Tactile graphics	Strong	Story	Weak / not primary
Portability	Medium	Medium; Dot Pad X is more tablet-like	High
Local manufacturing	Low	Low	Medium, India-origin product
Repairability	Low to medium	Low to medium	Medium
Open-source scalability	Limited	Limited	Limited
Best suited for	Premium institutions	Premium accessibility labs / institutions	Individual Braille readers

Funded by ciena | nasscom foundation

Proposed solution:

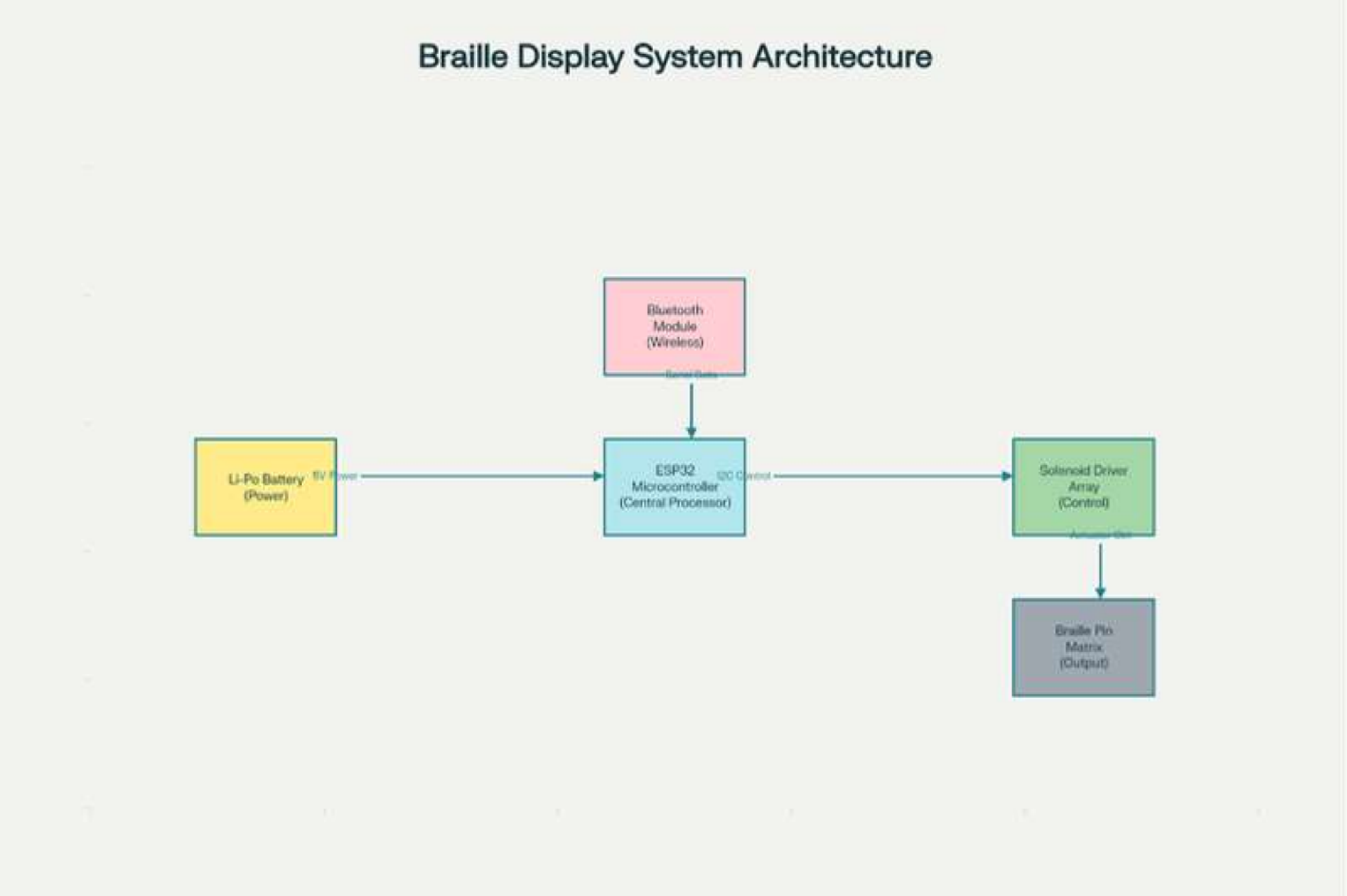
A lightweight, modular refreshable Braille display using 5V solenoids, ESP32 microcontroller, I2C architecture, and 3D-printed housing, reducing cost from $10,000–$20,000 to $500–$1,500.

It directly solves the affordability, multi-line display, and portability gaps—making Braille and STEM content accessible to millions of visually impaired learners and professionals at a fraction of the cost.

Combines low-cost components, open-source scalability, and local 3D printing, enabling rapid manufacturing, local repair, and 19x wider accessibility compared to current products.

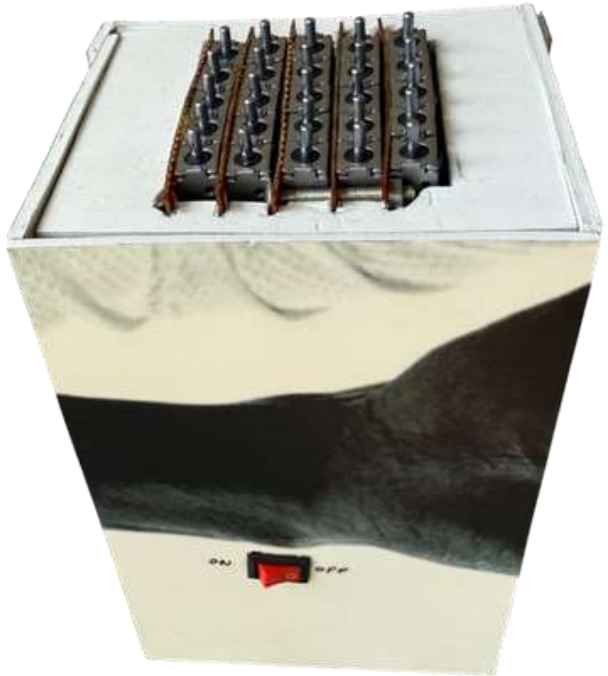
Our product uses intelligent software-hardware integration to empower visually challenged people in STEM Education

System Architecture

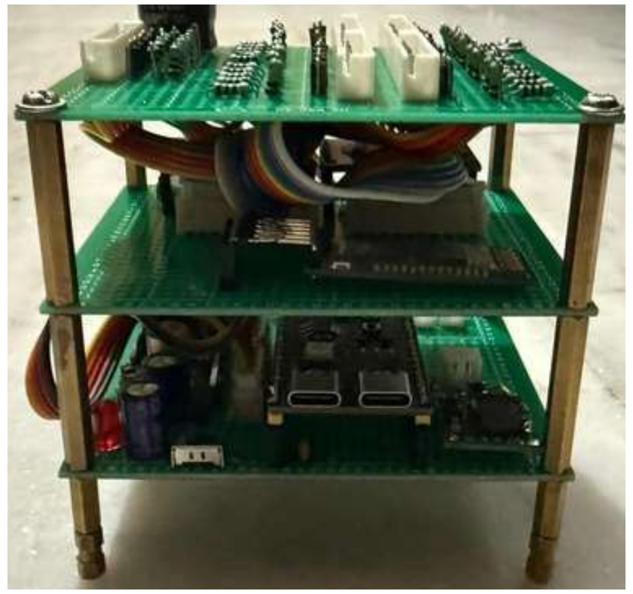


Bluetooth-connected app drives a tactile pin-matrix via ESP32 + solenoid drivers.

Hardware Prototype



Final Prototype



Internal Circuitry

VIISAR

Product & Prototype Readiness

Funded by ciena | nasscom foundation

SOFTWARE SERVICES

- Interactive Learning Modules AI-
- Powered Feedback System Portable
- Tactile-Compatible Interface
- Multilingual Support

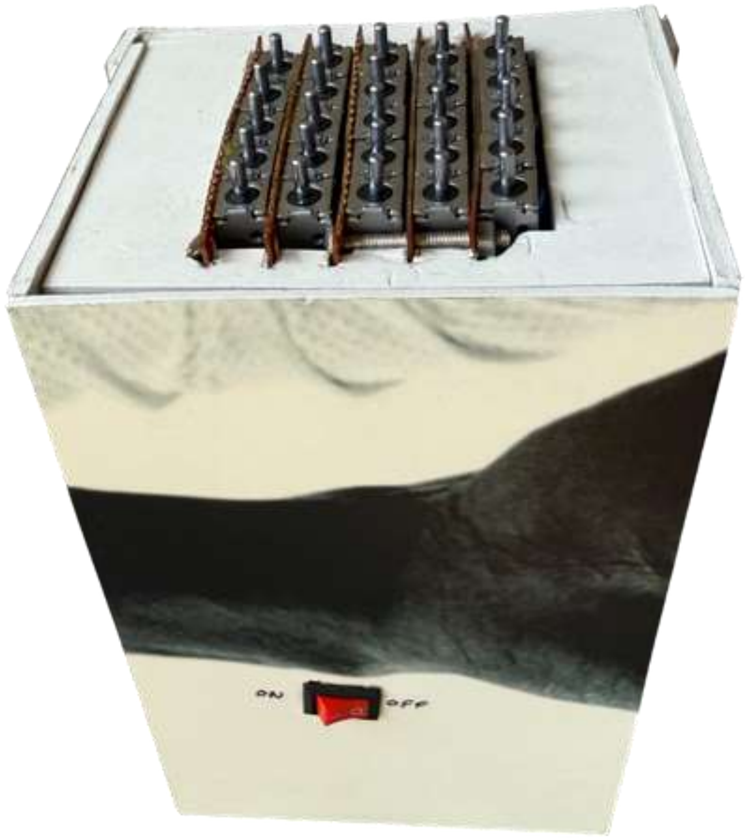
Website Link: https://viisar.vercel.app/

User Validation & Confirmation

Funded by ciena | nasscom foundation

Based on the feedback we received, we redesigned the hardware and made it smaller in size. This allows users to easily feel the images using their palm, and the device can now comfortably fit in the hand.

Before



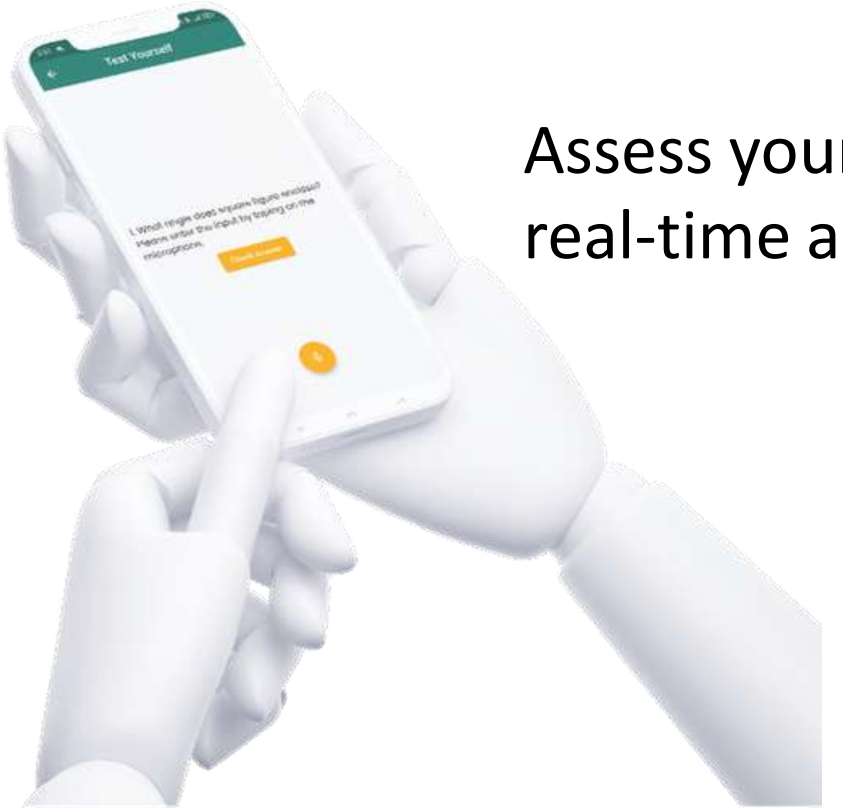
After

VIISAR

Product & Prototype Readiness

Funded by ciena | nasscom foundation

Use a Bluetooth-connected app and hardware to visually trace and comprehend figure geometry in real time



Assess your knowledge using real-time audio responses.

Key Features

Mobile app with voice read-aloud feature offering a diverse range of STEM study materials.

Explore and Learn various Maths and Science Concepts

VIISAR

Product Journey Experience

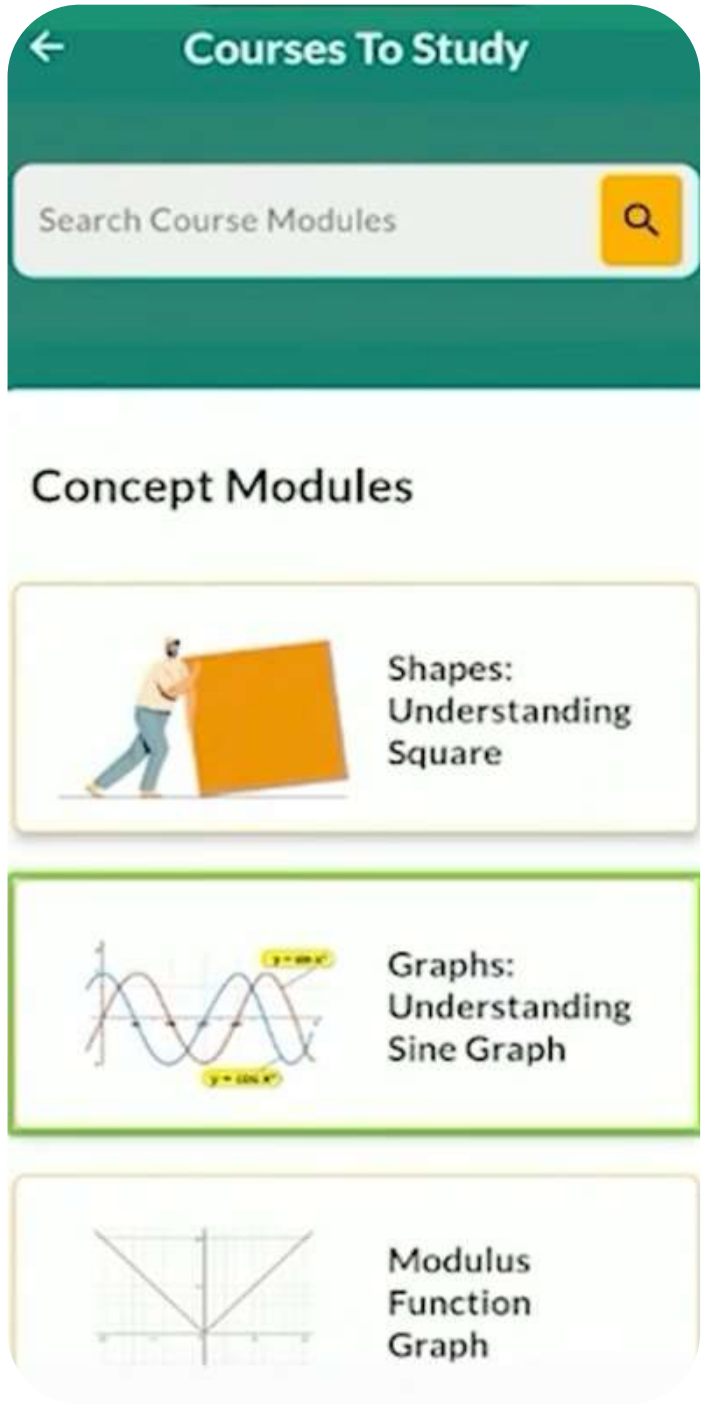
Funded by ciena | nasscom foundation

Screens

Starting with the app

Available options to learn, practice and test

Offering a diverse range of STEM study materials.



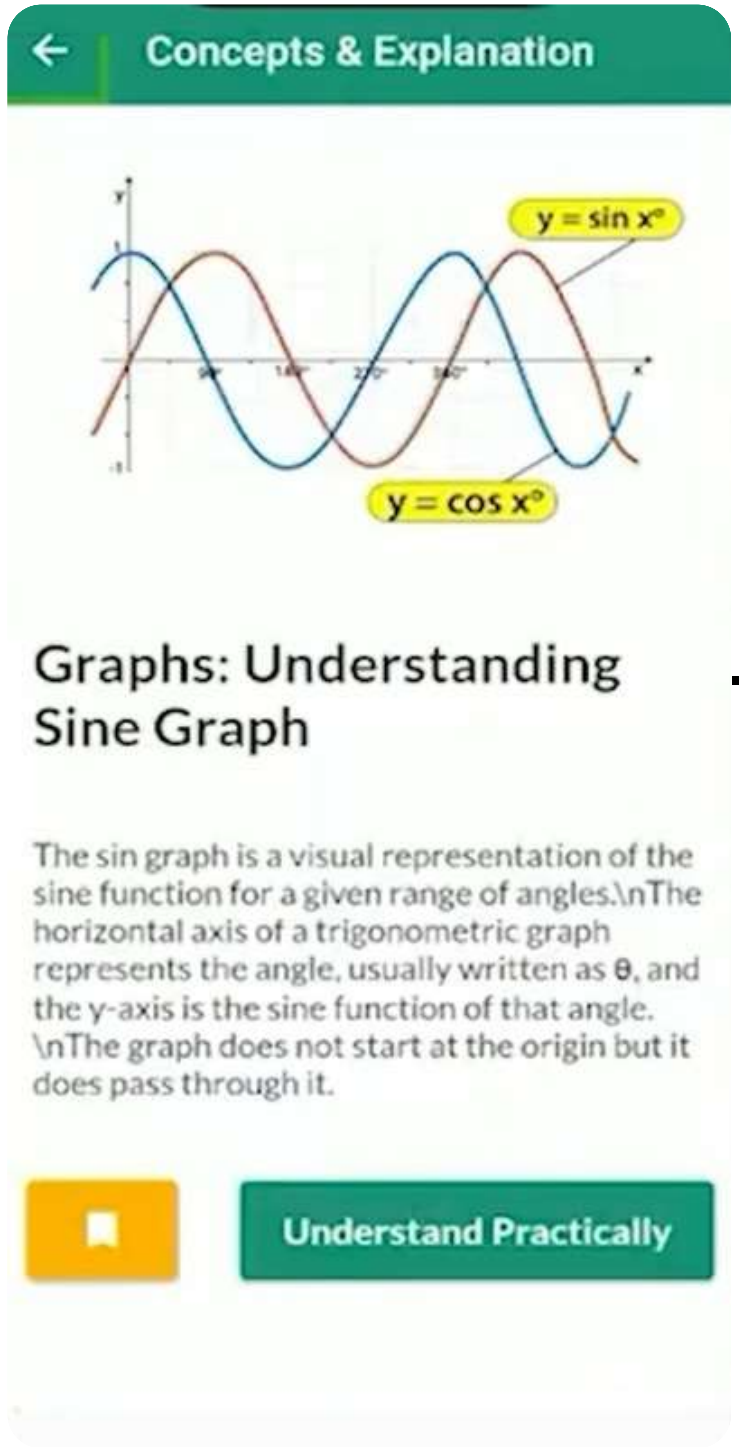
Explore and Learn various Maths and Science Concepts

VIISAR

Product Journey Experience

Funded by ciena | nasscom foundation

Screens



Understanding Sine Graph

Bluetooth enable option for connecting to hardware

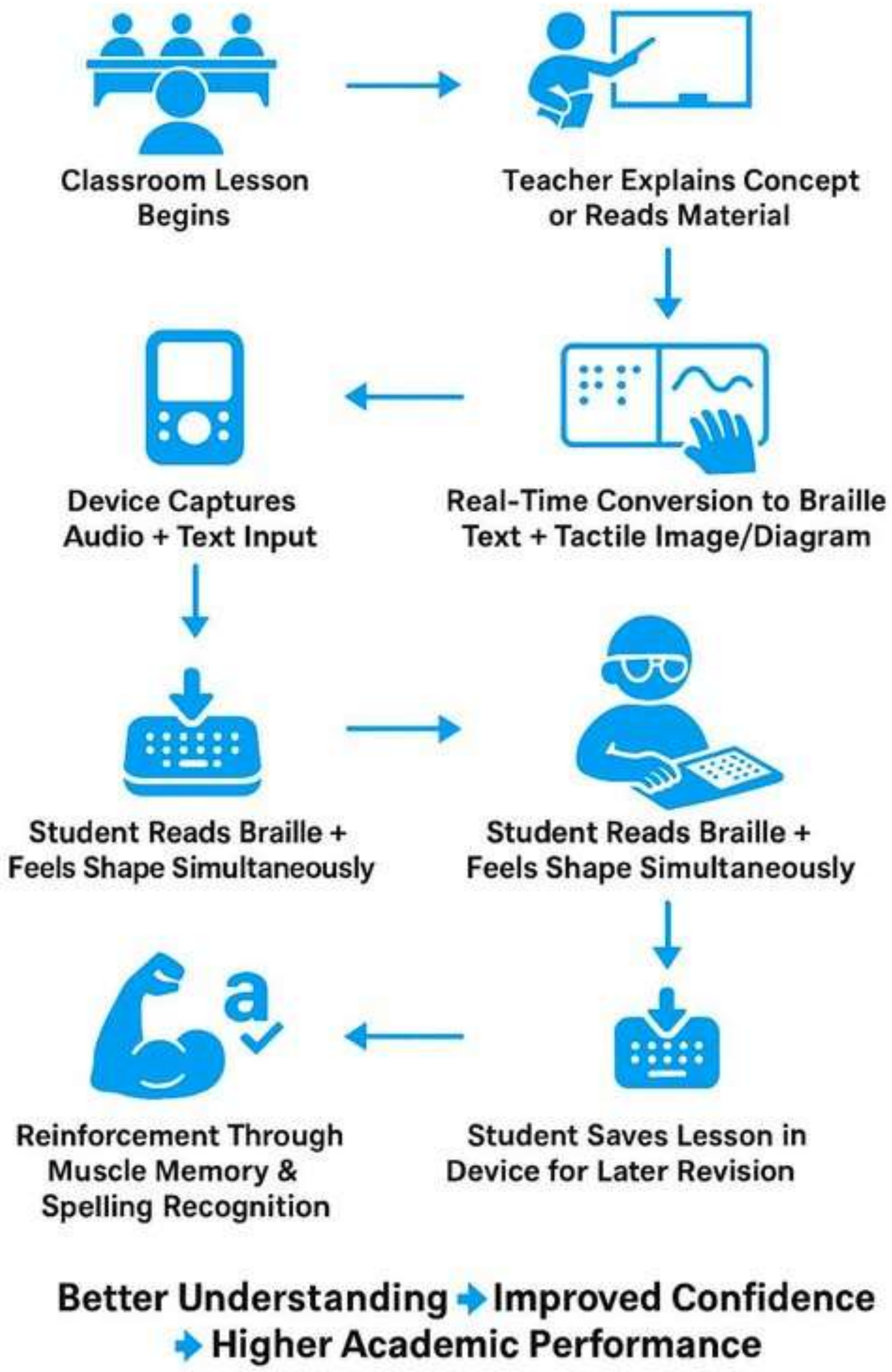
Hardware showing the practical representation of Sine Graph

VIISAR

Funded by ciena | nasscom foundation

User Journey

Product Journey Experience



Friction Points Identified:

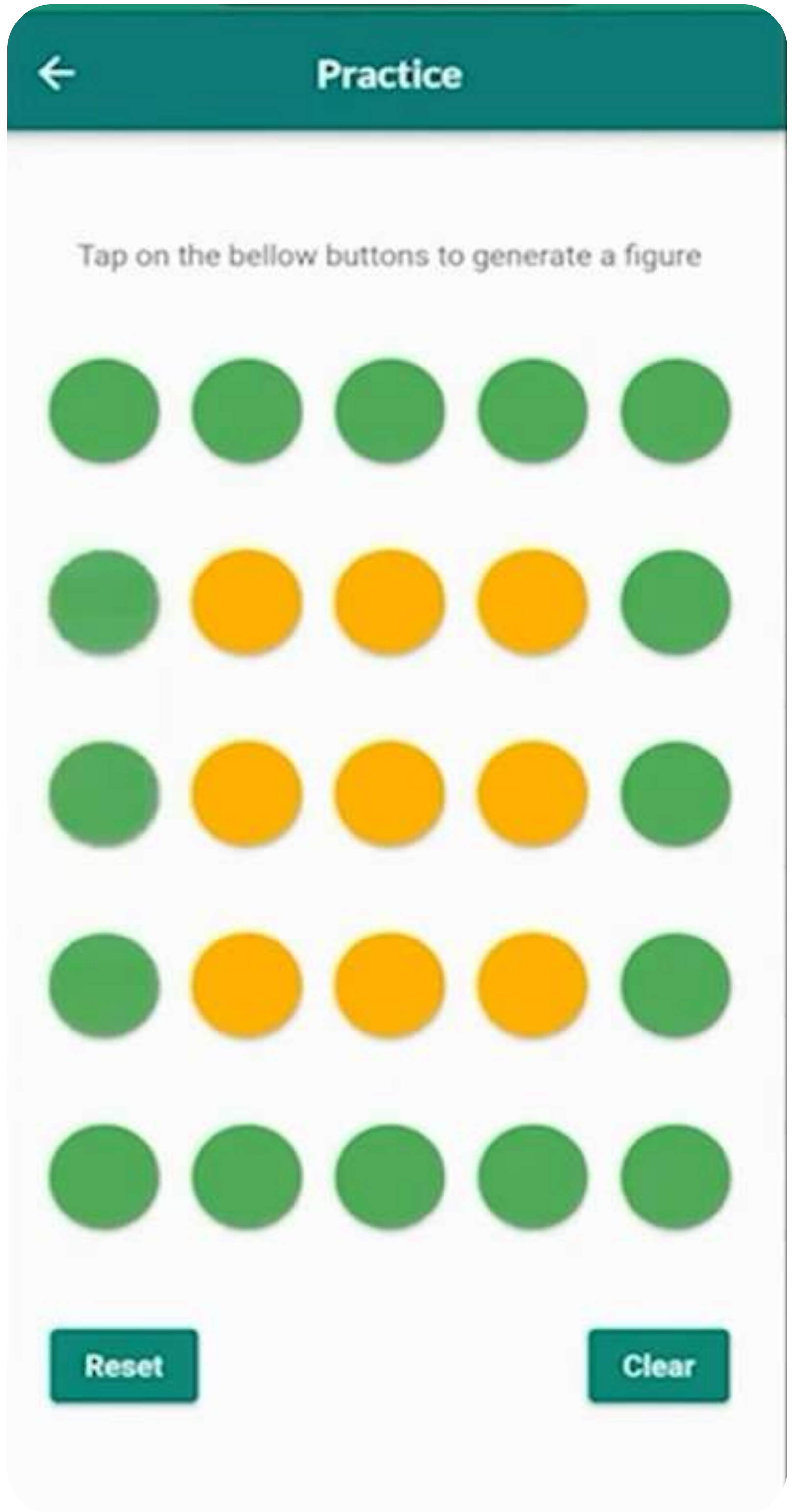
- In app, we are currently using talkback feature, but we need to implement voice controlled application.
- In hardware, a lot of heat is produced due to solenoids, so we need to reduce the heat that is being produced.

Product Journey Experience

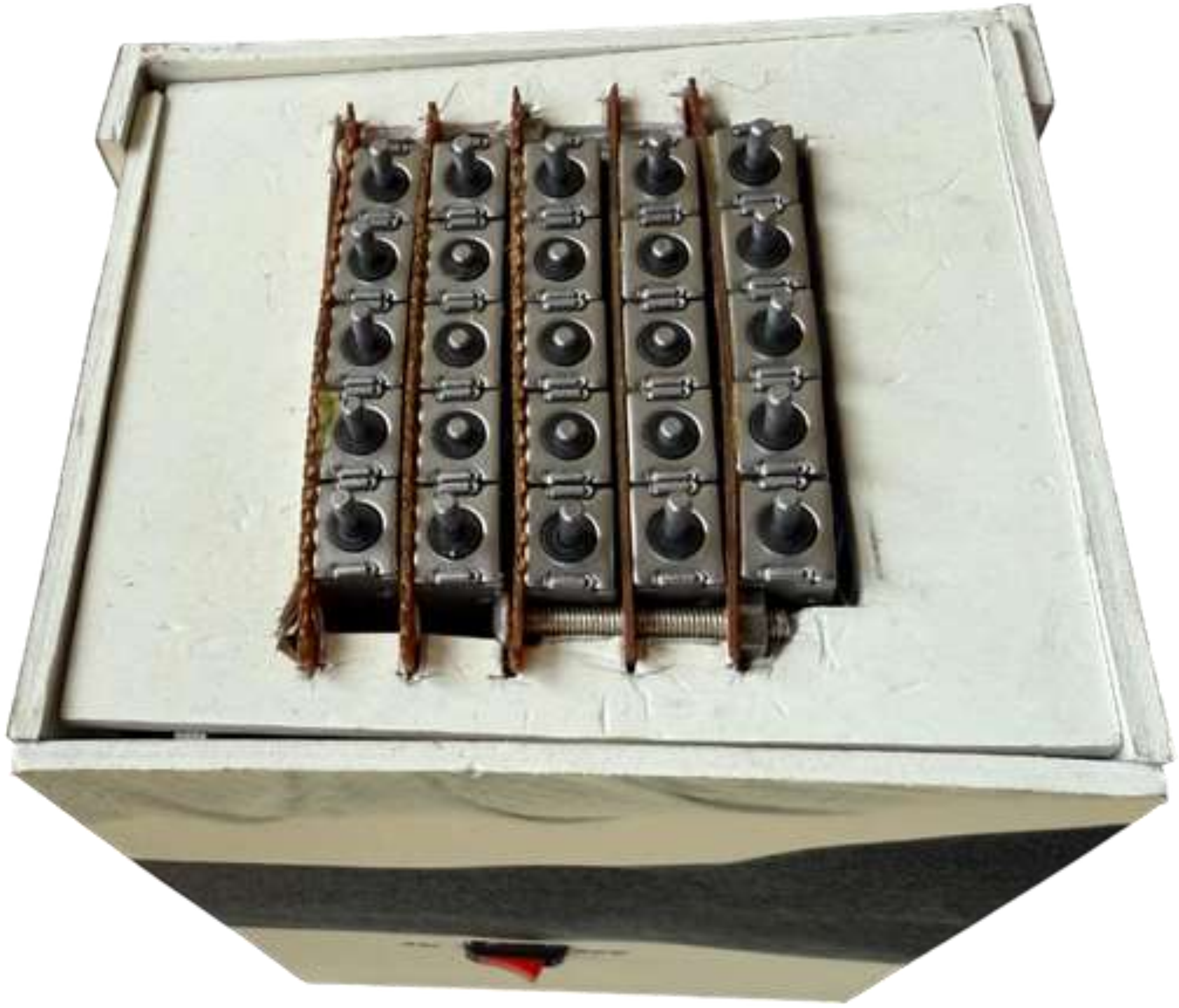
Funded by ciena | nasscom foundation

Screens

Connecting Bluetooth



Drawing square on app

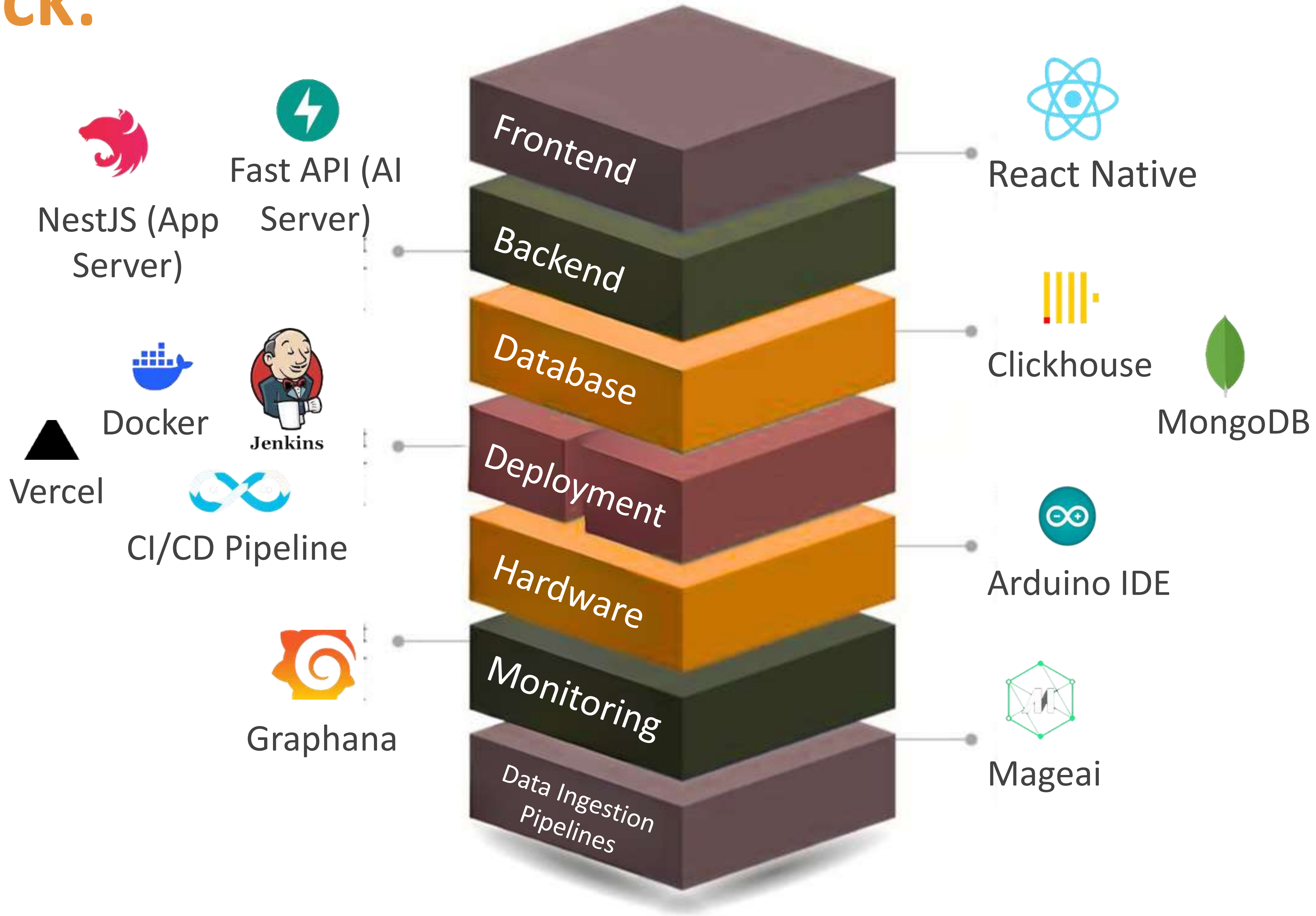


Hardware showing square figure

Product Demo & Technology

Funded by ciena | nasscom foundation

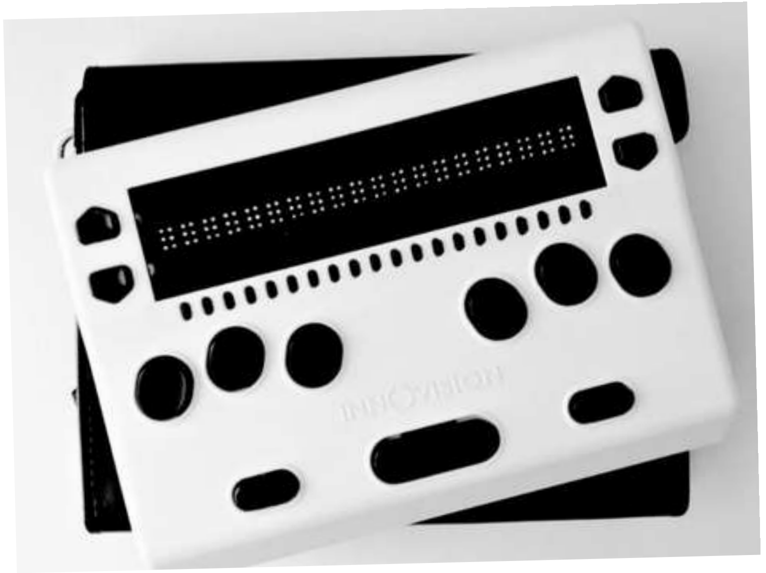
Tech Stack:



Innovation & Competitive Advantage

Funded by ciena | nasscom foundation

	Portability	Repairability	Cost-effective
VIISAR	✓	✓	✓
Graphiti	✗	✗	✗
Dot Pad	✗	✗	✗
BrailleMe (Innovision Tech)	✓	✗	✗



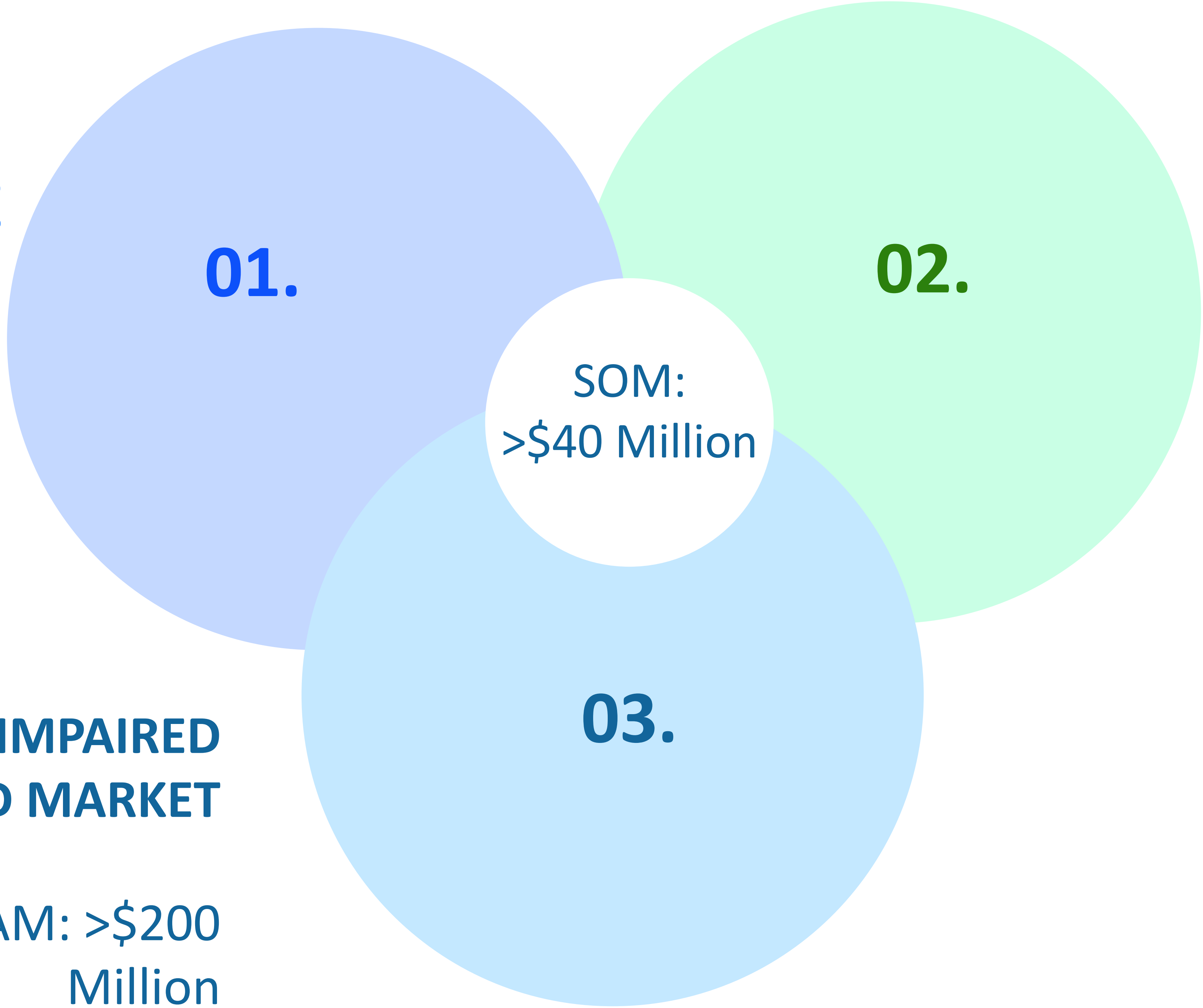
VIISAR

Market Opportunity & Business Model

Funded by ciena | nasscom foundation

WORLDWIDE VISUAL IMPAIRED MARKET

: > $2.2 Billion



WORLDWIDE BLIND MARKET

TAM: >$20 Billion

INDIAN VISUALLY IMPAIRED & BLIND MARKET

SAM: >$200 Million

VIISAR

Business Model Canvas

Funded by ciena

nasscom foundation

KEY PARTNERS

- Blind educational institutions
  - Sai Netraheen Sewa Sanstha, New Delhi
  - National Blind School, New Delhi
- Existing companies & startups
- NGOs
  - GirlUpFoundation, NYC
  - Voice of Specially Abled People

KEY ACTIVITIES

- Maintenance
- Hardware Development
- Software Development (mobile and web application for visually impaired)
- Sales & Marketing

KEY RESOURCES

- Software mobile application
- Cloud services
- Hardware manufacturing
- Intellectual resources
- Human resources

VALUE PROPOSITIONS

- Easy to understand STEM concepts.
- Easy to carry and use hardware.
- Hardware devices compatible with all devices.
- Providing an alternative to old methods of learning STEM concepts through Braille/interpreters/ abacus.
- Cost-effective.
- Increasing literacy and employability rate.

CUSTOMER RELATIONSHIPS

- Best-in-usage and services relationship.
- Budget-friendly.
- Feedback to improve customer experience.

CHANNELS

- Organize hackathons and workshops for visibility.
- Form partnerships with educational institutions and NGOs.
- Offer live demonstrations and ongoing support for customers.

CUSTOMER SEGMENTS

- Visually impaired people (any age group and people suffering through any type of visual impairment)
- Blind Educational Institutions and Organizations
- Blind associations
- B2B

COST STRUCTURE

- Manufacturing of Hardware
- Advertisement
- Software platform charges
- Dealerships
- Maintenance of hardware
- Offline campaigns
- Employees

REVENUE STREAMS

- Average number of blind people using our software per year: 500
- Average software charges: INR 300 per month
- Hardware charges (one-time): INR 12000/ unit
- Average number of hardware units sold: 200 units
- Total income per year: 500*(300*12)+12000*200= 34.5 lakhs
- After removing expenditure, estimated profit = 10 lakhs per annum

VIISAR

Social Impact & Schedule

Funded by ciena nasscom foundation

Who is Benefited?

Primary - Visually impaired / blind students (especially in under-resourced schools)

Secondary - Teachers, special educators, schools, parents/caregivers, NGOs working in disability inclusion

Real World Use Case:

Teacher introduces topic (e.g., triangles / graphs / coordinate geometry)

Student uses the tool to feel the diagram (tactile output) + hear guidance (audio)

Student solves practice questions independently, gets feedback

VIISAR

Social Impact & Schedule

Funded by ciena | nasscom foundation

SDGs	Direct Impact	Indirect Impact
4 QUALITY EDUCATION	Accessible STEM via tactile + audio learning; improves concept understanding and confidence	Teacher enablement + curriculum integration for inclusive STEM classrooms and assessments.
8 DECENT WORK AND ECONOMIC GROWTH	STEM readiness → pathways to higher studies, skills development, and employability.	Long-term economic inclusion by reducing dependency and widening career opportunities.
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	Affordable assistive-tech learning infrastructure (device + app ecosystem) for learners.	Modular design supports scalable manufacturing, local repair, and sustainable deployment.
10 REDUCED INEQUALITIES	Cuts disability-based learning barriers; supports equal participation without constant assistance.	Scales access through schools/NGOs to underserved communities and low-income learners.

Estimated Scale

Pilot target (first deployment)
3–5 schools / 50–150 students

Post-pilot (after iteration)
10–20 schools / 300–800 students

Measurable outcomes

- **Learning gain:** pre/post concept test improvement (%)
- **Independence:** % tasks completed without assistance
- **Time efficiency:** time to complete diagram-based problems
- **Teacher adoption:** weekly class sessions using tool
- **Retention rate:** attendance or participation proxy
- **Cost proxy:** reduction in material dependence per learner

VIISAR

Why We Should Be Selected

Funded by ciena | nasscom foundation

Key reasons that make VIISAR the right choice

Proven Impact

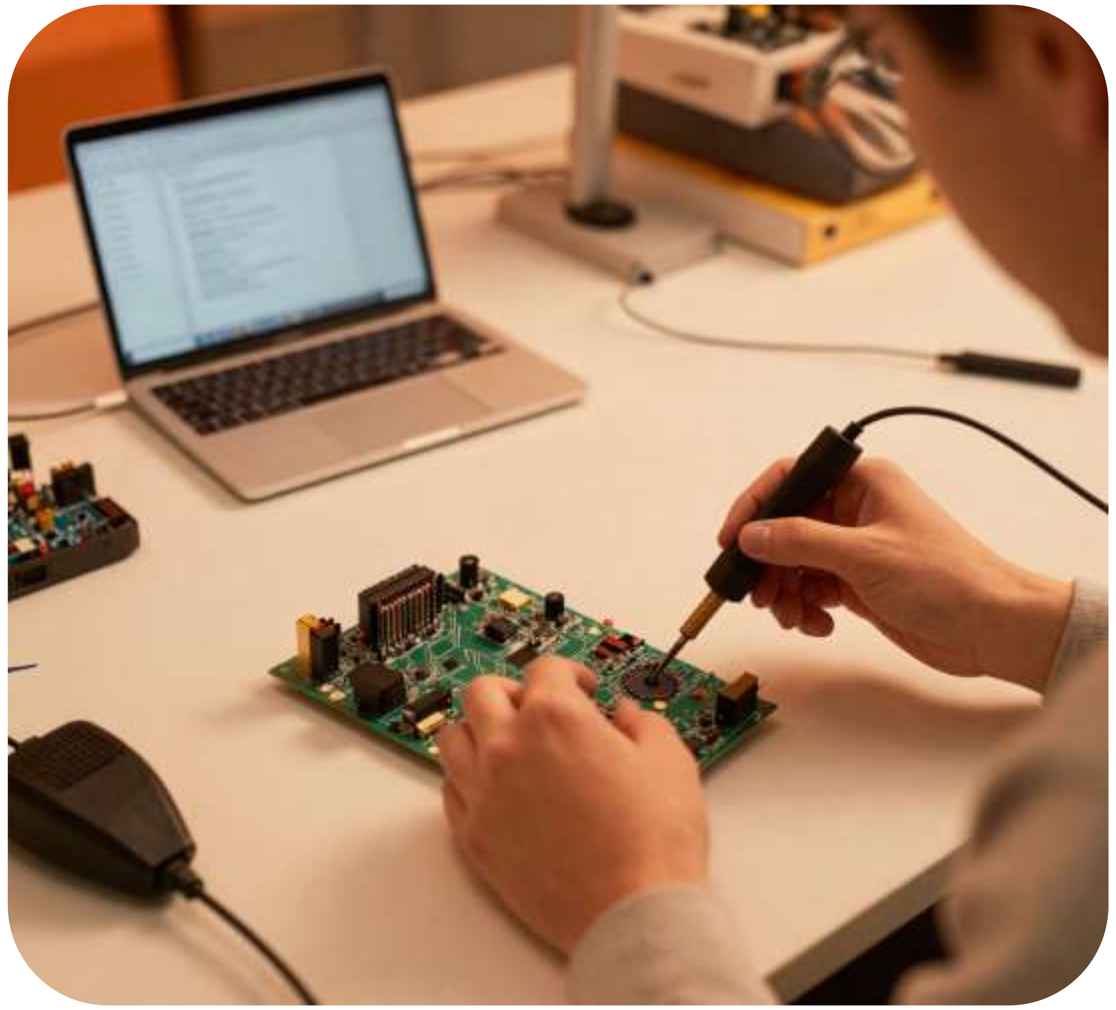
Our solution directly addresses the learning gap for 39M+ visually impaired individuals, with measurable outcomes in pilot testing.

Scalable & Affordable

Modular hardware + mobile app architecture enables rapid deployment at <$1000/unit, making it accessible for NGOs and schools.

Strong Execution Team

Cross-functional team with deep expertise in assistive tech, embedded systems, and inclusive education design.



39M+
Lives Impacted

95%
Pilot Satisfaction

<$1000
Per Unit Cost

6 Mo
To Deploy

VIISAR

Funded by ciena | nasscom foundation

Thank You